

Practice: Evaluation Research

Session 02 - MLM and CSM

psy112 - Evaluation Research

Faculty VI / UOL

Summer term 2026

Seminar content - Practice (P)

- 09.04. - Th: Introduction to evaluation research
- 16.04. - P: **AN(C)OVA**
- 23.04. - Th: Modeling change at the level of observed variables
- 30.04. - P: **Multilevel and SEM change score models**
- 07.05. - Th: Modeling change with growth curves
- 14.05. - **Public holiday**
- 21.05. - Th: Modeling change at the level of latent variables
- 28.05. - P: **Multilevel and SEM change score models**
- 04.06. - Th: **Causality**
- 11.06. - P: **Latent change score and growth curve models**
- 18.06. - Th: **Sampling and matching**
- 25.06. - P: **Propensity score matching**
- 02.07. - Th: **Systematic reviews and metaanalysis**
- 09.07. - P: **Open topic**

Content

- 1 Mixed ANOVA
- 2 Multilevel Models of Change
- 3 Change Score Model

Mixed ANOVA

- **Core Idea:** Mixed ANOVA is used for research designs that include **both**
 - ▶ a **Between-Person Factor**: Comparing two or more *independent* groups of participants (e.g., treatment vs. control group)
 - ▶ a **Within-Person Factor**: Measuring the *same* participants multiple times or under multiple conditions (e.g., pre-test vs. post-test; this is the repeated measures part)
- **Purpose:** It allows to simultaneously estimate
 - ▶ differences *between* the groups (main effect of the between-person factor)
 - ▶ changes *within* individuals over time/conditions (main effect of the within-person factor)
 - ▶ the **interaction** between these factors: Does the within-person change differ depending on which group a person belongs to?

Content

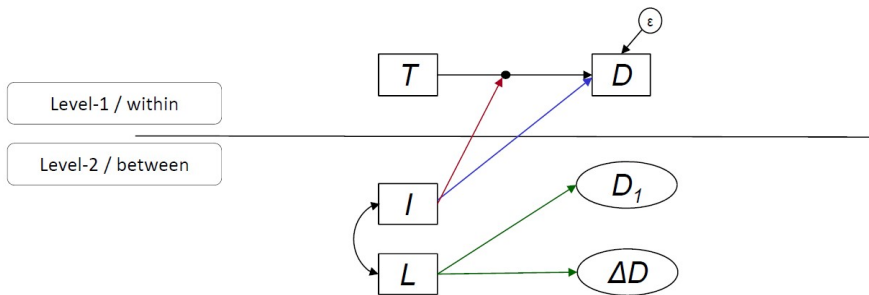
1 Mixed ANOVA

2 Multilevel Models of Change

3 Change Score Model

Multilevel model

- Between-person IV: Intervention (I) no/yes = 0/1
- Within-person IV : Time (T) pre/post = 0/1
- DV: Depression (D)
- Covariate: Loneliness (L)



Exercise: Write the equation for the multilevel regression model and interpret the 5 regression coefficients (colored + black) estimated in the model.

Multilevel Model Equation

This model predicts Depression score (D) based on Time (T), Intervention group (I), Loneliness (L), and their interactions.

Variables:

- D : Depression score (Level 1 DV)
- T : Time (Level 1 IV, coded 0=pre, 1=post)
- I : Intervention group (Level 2 IV, coded 0=control, 1=intervention)
- L : Loneliness (Level 2 Covariate)

Model Equation:

$$D_{ti} = b_0 + b_1 \cdot T_{ti} + b_2 \cdot I_i + b_3 \cdot (T_{ti} \cdot I_i) + b_4 \cdot L_i + b_5 \cdot (T_{ti} \cdot L_i) + \epsilon_{ti}$$

Interpretation of Coefficients

$$D_{ti} = b_0 + b_1 \cdot T_{ti} + b_2 \cdot I_i + b_3 \cdot (T_{ti} \cdot I_i) + b_4 \cdot L_i + b_5 \cdot (T_{ti} \cdot L_i) + \epsilon_{ti}$$

b_0 : Intercept

Path: Baseline (no arrow in the graph)

Interpretation: Estimated average D at pre-test ($T = 0$) for the control group ($I = 0$) when Loneliness is 0 ($L = 0$, needs to be centered)

b_1 : Main Effect of Time T

Path: Black arrow ($T \rightarrow D$), slope for ref. group

Interpretation: Estimated average change in D from pre- to post-test ($T = 0$ to $T = 1$) for the control group ($I = 0$) when Loneliness is 0 ($L = 0$)

b_2 : Main Effect of Intervention I

Path: Blue arrow ($I \rightarrow D$)

Interpretation: Estimated average difference in D between intervention ($I = 1$) and control ($I = 0$) groups at pre-test ($T = 0$), holding L constant

Interpretation of Coefficients

$$D_{ti} = b_0 + b_1 \cdot T_{ti} + b_2 \cdot I_i + b_3 \cdot (T_{ti} \cdot I_i) + b_4 \cdot L_i + b_5 \cdot (T_{ti} \cdot L_i) + \epsilon_{ti}$$

b_3 : Interaction Effect $T \times I$

Path: Red arrow (I modifying $T \rightarrow D$)

Interpretation: Estimates how much the change in D over time differs for the intervention group ($I = 1$) compared to the control group ($I = 0$), holding L constant

b_4 : Main Effect of Loneliness L

Path: Green arrow (top) ($L \rightarrow D$)

Interpretation: Estimated difference in D at pre-test ($T = 0$) associated with a one-unit increase in L , holding I constant

b_5 : Interaction Effect $T \times L$

Path: Green arrow (bottom) (L modifying $T \rightarrow D$)

Interpretation: Estimates how much the change in D over time differs for each one-unit increase in L , holding I constant

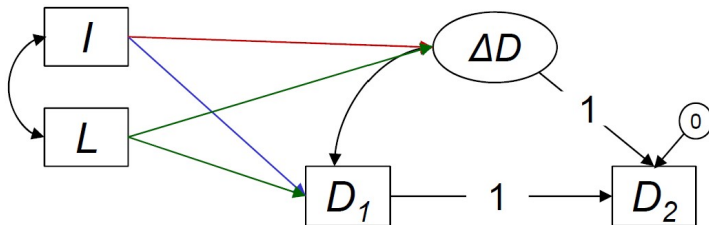
Note: The curved arrow between I and L represents their correlation, not a coefficient in this model predicting D .

Content

- 1 Mixed ANOVA
- 2 Multilevel Models of Change
- 3 Change Score Model

Change (or Difference) Score Model (Path modeling)

- Between-person IV: Intervention (I) no/yes = 0/1
- Within-person IV : Time (T) pre/post = 0/1
- DV: Depression (D)
- Covariate: Loneliness (L)



Exercise: Write the path model equations and interpret the 4 regression coefficients (colored) estimated in the model.

Change Score Path Model: Equations

This path model estimates the change (ΔD) between baseline depression (D_1 , $T=0$) and (D_2 , $T=1$). Given the model constraints (paths fixed to 1), the change score is defined as $\Delta D = D_2 - D_1$.

Variables:

- D_1 : Depression score at Time 0 (Pre-test)
- D_2 : Depression score at Time 1 (Post-test)
- ΔD : Change in Depression ($D_2 - D_1$)
- I : Intervention group (0=control, 1=intervention)
- L : Loneliness (Covariate)

Path Model Equations:

$$D_{1i} = b_{0D1} + b_1 \cdot I_i + b_2 \cdot L_i + \epsilon_{D1i}$$

$$\Delta D_i = b_{0\Delta D} + b_3 \cdot I_i + b_4 \cdot L_i + \epsilon_{\Delta Di}$$

$$D_{2i} = D_{1i} + \Delta D_i \quad (\text{Implied by diagram})$$

Where i denotes the individual, b_1, b_2, b_3, b_4 correspond to the colored paths, and $b_{0D1}, b_{0\Delta D}$ are intercepts.

Interpretation of Path Coefficients

$$D_{1i} = b_{0D1} + b_1 \cdot I_i + b_2 \cdot L_i + \epsilon_{D1i}$$
$$\Delta D_i = b_{0\Delta D} + b_3 \cdot I_i + b_4 \cdot L_i + \epsilon_{\Delta D i}$$

b_1 : **Effect of I on D_1**

Path: Blue arrow ($I \rightarrow D_1$)

Interpretation: Estimated average difference in baseline depression (D_1) between intervention ($I = 1$) and control ($I = 0$) groups, holding L constant

b_2 : **Effect of L on D_1**

Path: Green arrow (bottom) ($L \rightarrow D_1$)

Interpretation: Estimated average difference in baseline depression (D_1) for a one-unit increase in L , holding I constant

Interpretation of Path Coefficients

$$D_{1i} = b_{0D1} + b_1 \cdot I_i + b_2 \cdot L_i + \epsilon_{D1i}$$
$$\Delta D_i = b_{0\Delta D} + b_3 \cdot I_i + b_4 \cdot L_i + \epsilon_{\Delta D i}$$

b_3 : Effect of I on ΔD

Path: Red arrow ($I \rightarrow \Delta D$)

Interpretation: Estimated average difference in the *change* in depression (ΔD) between intervention ($I = 1$) and control ($I = 0$) groups, holding L constant

b_4 : Effect of L on ΔD

Path: Green arrow (top) ($L \rightarrow \Delta D$).

Interpretation: Estimated average difference in the *change* in depression (ΔD) for a one-unit increase in L , holding I constant

Note: The curved arrow between I and L represents their correlation. The black arrows define $\Delta D = D_2 - D_1$.

Next session

Topic: Latent Change Score Model

©Prof. Andrea Hildebrandt & Dr. Daniel Kristanto

andrea.hildebrandt@uni-oldenburg.de